

Angela Qian

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EDUCATION

Purdue University | *B.S. in Computer Science, Minor in Mathematics*

Aug 2023 – May 2027

EXPERIENCE

Undergraduate Research Assistant | *CAMP Lab at Purdue University*

Feb 2025 - Present

- Co-authored a paper submitted to NeurIPS on Bayesian social deduction with graph-informed language models
- Helped conduct experiments involving human participants
- Assisted in developing and evaluating methods for hallucination detection

Software Engineering Intern | *Cepton Technologies Inc.*

May 2025 - Aug 2025

- Built in Rust a real-time object tracking pipeline using LiDAR-generated point cloud data on a per-frame basis
- Developed a 2D rectangle selection tool for a 3D point cloud viewer, enabling users to select points from any camera perspective and display their coordinates dynamically by transforming 3D point data using projection matrices
- Helped develop a neural network with U-Net segmentation to do real-time foreground-background segmentation masking, given depth and reflectivity maps

Teaching Assistant | *Purdue University Computer Science Department*

Jan 2025 - May 2025

- Led weekly PSO sessions, teaching problem-solving techniques and reinforcing theory taught in class
- Graded assignments, providing detailed feedback to help students improve and better understand material
- Held weekly office hours to clarify course material and guide students through homework challenges
- Conducted academic dishonesty investigations and interviewed students in coordination with course staff

Undergraduate Researcher | *Purdue University Vertically Integrated Projects*

Aug 2024 – Dec 2024

- Designed a program that uses convolution, the Laplacian filter, and the Sobel filter to detect image blur
- Developed algorithms involving Fourier Transforms to generate clear, focused photos from blurred images
- Presented in Purdue's 2024 Fall Undergraduate Research Expo

Competition Mathematics Instructor | *AlphaStar Academy*

July 2024 – Aug 2024

- Delivered lectures on Algebra, Number Theory, Geometry, and Counting/Probability
- Held review sessions with practice problems to prepare students for the AMC10, AMC12, and AIME exams
- Presented feedback to students and parents, highlighting focus areas for continued academic growth

PUBLICATIONS

Bayesian Social Deduction with Graph-Informed Language Models

Shahab Rahimirad, Guven Gergerli, Lucia Romero, **Angela Qian**, Matthew Lyle Olson, Simon Stepputtis, Joseph Campbell (2025) *In submission for NeurIPS 2025*

PROJECTS

2048 AI Player | *PyTorch, TensorFlow, Jupyter Notebook, JavaScript*

June 2025 - Present

- Developed a neural network to play 2048 using imitation learning trained on a custom dataset of my own gameplay
- Built model in PyTorch to predict actions from board states, and exported it via ONNX to deploy in TensorFlow.js
- Currently extending the project with reinforcement learning to fine-tune decision making through self-play

ScribbleScore Handwriting Improvement | *Python, OpenCV, Numpy, React, Flask*

Feb 2025

- Worked in a group of 4 over the course of a 36-hour hackathon (Boilermake XII)
- Designed bounding-box logic to group words and measure spacing consistency across clusters
- Implemented Sobel edge detection, line of best fit analysis to assess writing straightness, extracting text from images, and computing score

SKILLS

Languages: English (Native), Mandarin Chinese (Native)

Programming: Python, Rust, C/C++, JavaScript, Typescript, x86_64 Assembly, SQL, MongoDB, HTML, CSS, Java

Developer Tools: Git, GitHub, VS Code, Eclipse, Jupyter Notebook, Linux, Overleaf, React, LaTeX, VIM, MySQL, Neo4J, Next.js, TailwindCSS, Agile & Scrum

Libraries: OpenCV, Matplotlib, PyTorch, TorchVision, TensorFlow, Pandas, NumPy